

Assessing Drug-Induced lncRNA responses in cancer cell lines using the Tahoe-100M Atlas

Αγγελοπούλου Αθηνά

Τμήμα Φαρμακευτικής

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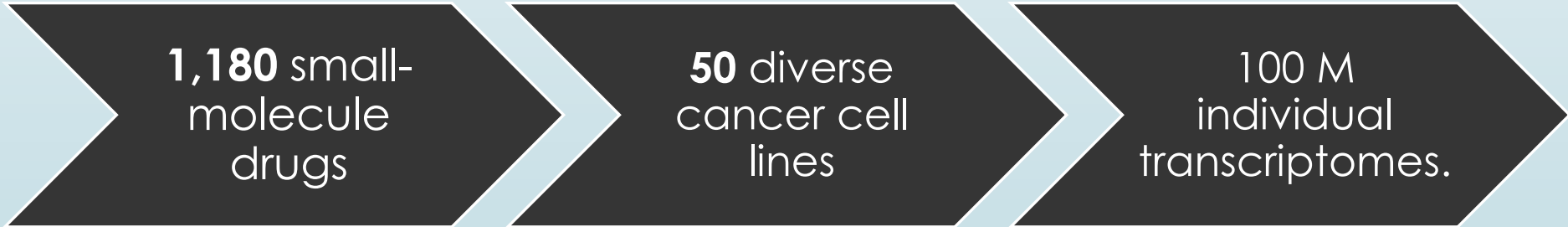
Overview of the basis paper:

Tahoe-100M: A Giga-Scale Single-Cell Perturbation Atlas for Context-Dependent Gene Function and Cellular Modeling

Published by Vevo Therapeutics and the Arc Institute

The "Big Idea": To build predictive "virtual cell" models, we need a massive, standardized map of how different cells react to different perturbations.

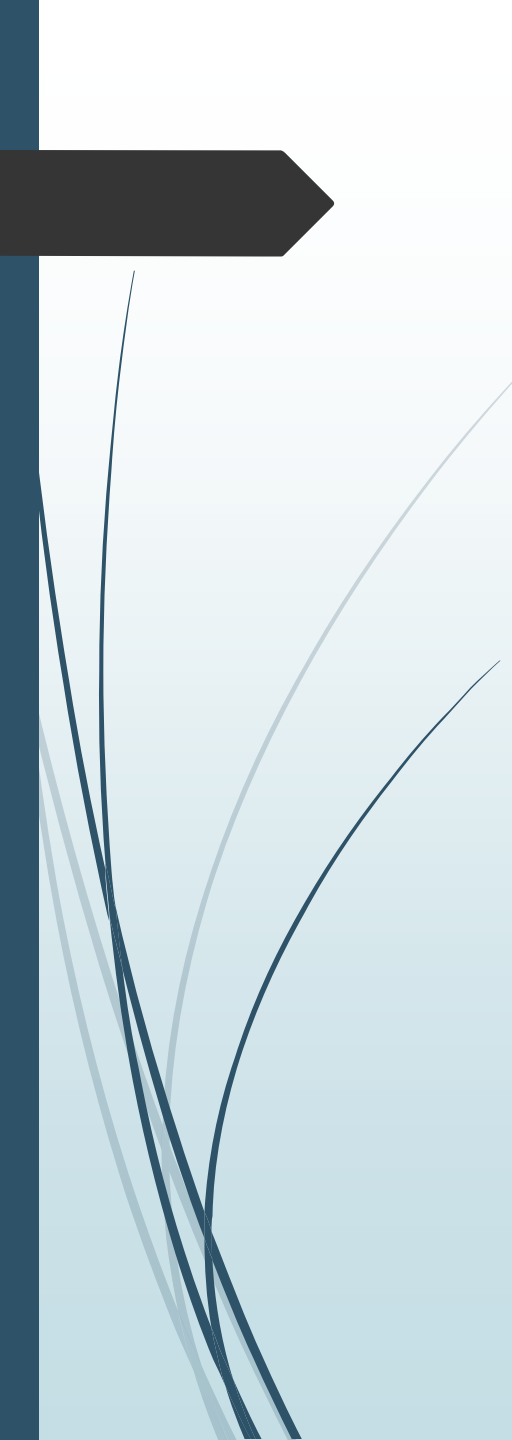
Tahoe-100M is **50x larger** than all previous public drug-perturbed datasets combined!



1,180 small-
molecule
drugs

50 diverse
cancer cell
lines

100 M
individual
transcriptomes.



Why do we need it ?

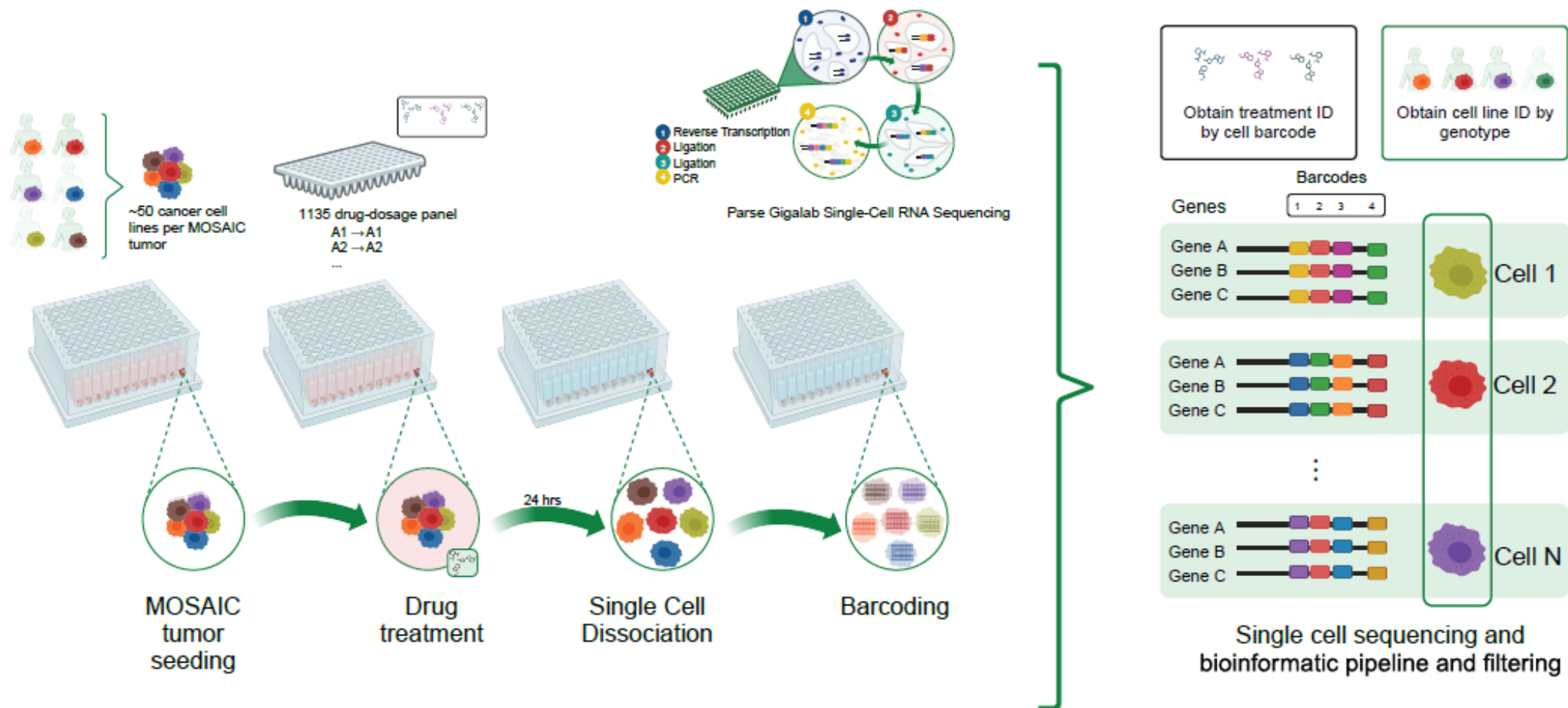
Problems:

- Bulk transcriptomic measurements often mask important heterogeneity (cell cycle phase, genetic background etc.) within cell populations.
- Limited availability of large-scale, perturbation-focused single-cell datasets.
- Aggregation of data from multiple studies has challenges such as batch effects, inconsistent protocols, and limited coverage of perturbations.

Solution: **The Solution: The Mosaic™ Platform**

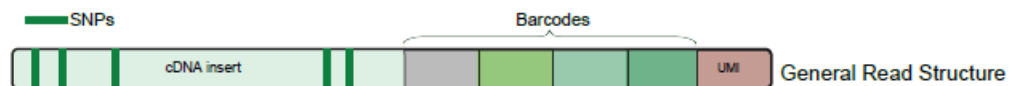
A

Tahoe-100M Data Generation



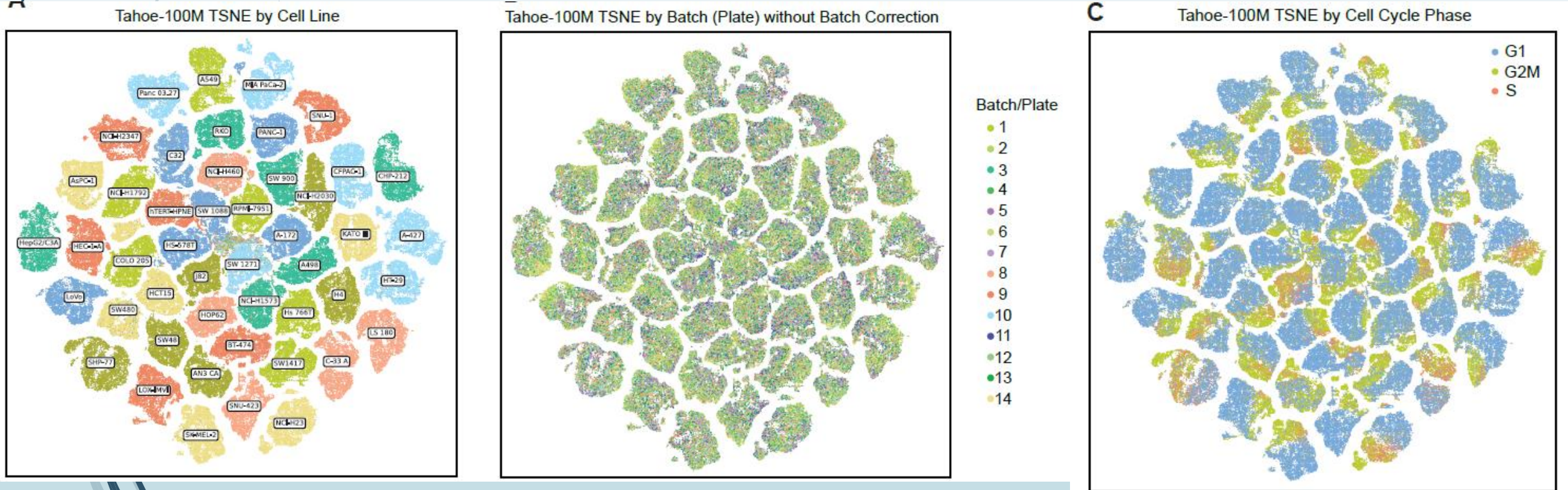
B

Read Structure Deconvolution



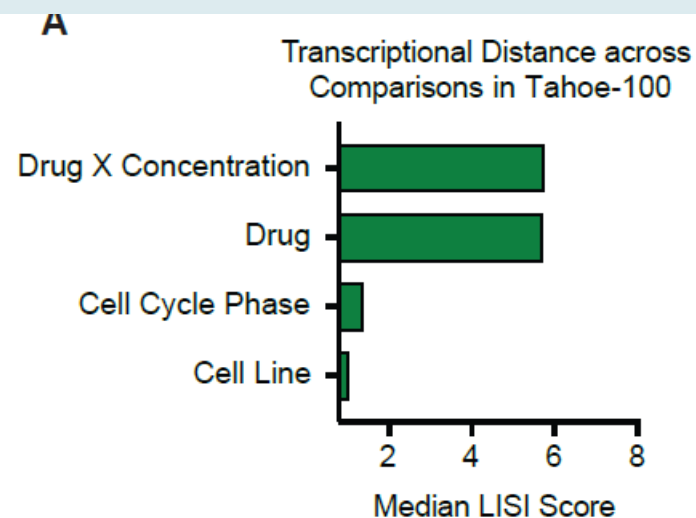
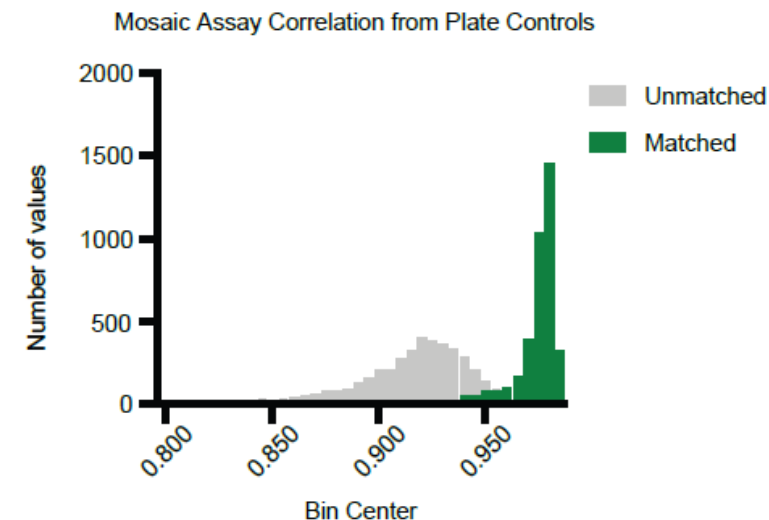
Results

- Dimensionality reduction with scVI to learn a 10-dimensional embedding for each cell.
- 2D-tSNE representation of the SCVI model, plotting a random selection of 200,000 cells with equal contribution from the top 47 cell lines by abundance.



Quality Control

- Pseudobulk grouping of all cells belonging to the same plate, cell line, and drug concentration treatment and calculation of the **Pearson correlation** between the two biological replicate 96-well plates in the Atlas .
- Higher correlation between conditions with matched treatment and cell identity compared to unmatched comparisons between the two plates.
- Estimation of **Local inverse Simpson's index (LISI)** based on different metadata factors over a subsample of the data. Cell line identity, followed by cell cycle phase, are the most major factors in driving stratification of the data.



Differential gene-set score calculations for dimensionality reduction

Tahoe

- ❑ Genes were filtered to the intersection of genes in the reference genome and those present in gene set collections h.all, c5.GO, c6.all, and c2.cp from **MSigDB**, resulting in 20,250 genes.
- Gene variability was estimated using 100,000 samples and calculating the standard deviation of the sample per gene. The top 50% most variable genes (**10,126 genes**) were selected.
- ❑ Gene sets were filtered by size (15–500 genes) and by overlap with at least 10 highly variable genes, resulting in **7,390 retained gene sets**.

My changes

- ❑ Genesets from **Reactome** and **GO biological processes**.
MSigDB + IncRNAlyzr
- ❑ **7685 sets, 18,507 unique genes, 3290 IncRNA genes.**
- ❑ After the filters:

Sets retained:	4527
Max HVGs found:	270
Min HVGs found:	10

Differential gene-set score calculations for dimensionality reduction

Tahoe

- ❑ Gene-set activity was computed using adapted **Vision scores**, defined as the log geometric mean of scVI-normalized expression values for genes within each set.
- ❑ Differential Vision scores were calculated by comparing drug-treated cells to matched DMSO controls within each plate and cell line, discarding comparisons with fewer than 50 cells and leaving **62,927 comparisons**
- ❑ Vision score = $\text{mean}(\log(\text{expr}(\text{gene}) + \epsilon))$ for all genes in the set)
- ❑ Differential Score = $\text{mean}(\text{scores of drug cells}) - \text{mean}(\text{scores of DMSO cells})$

My changes

- ❑ Comparisons 59,086
- ❑ Differential Vision score matrix:
 $59,086 \times 4,527$

Positive → the gene set is more active in drug-treated cells than DMSO

Negative → the gene set is less active in drug-treated cells than DMSO

Near zero → no meaningful difference

Differential gene-set score calculations for dimensionality reduction

Tahoe

- For each comparison, 250 Monte Carlo samples of normalized expression were generated with scVI, converted to gene-set scores, and summarized by the median across samples.
- Median differential gene-set scores were assembled into a matrix of 7,390 gene sets across 62,927 drug-cell line-plate comparisons.

My changes

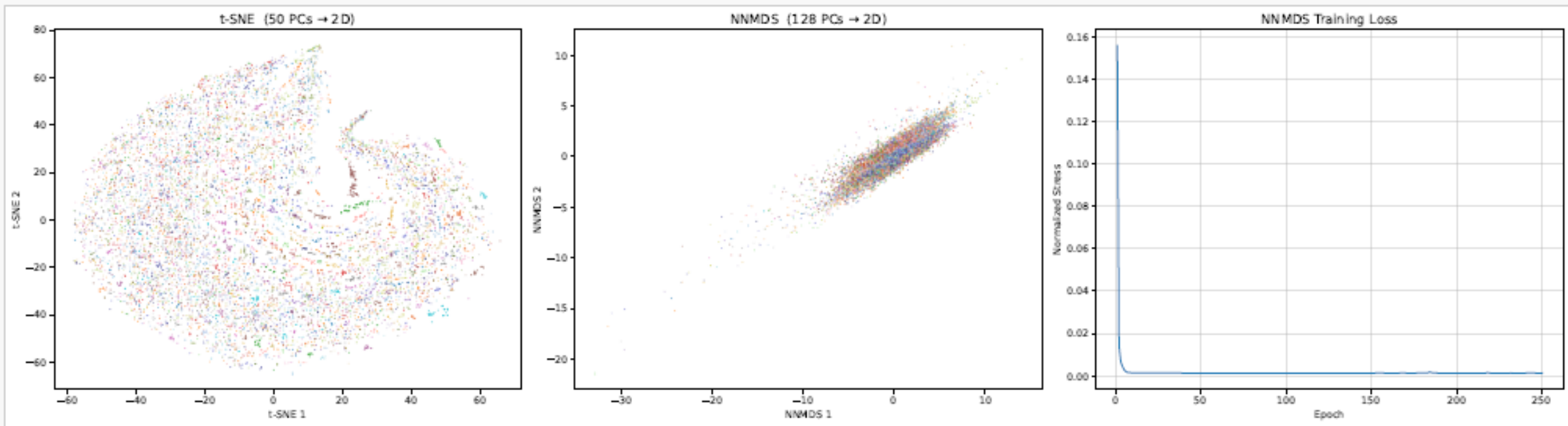
Problem: Too much space or too much time!

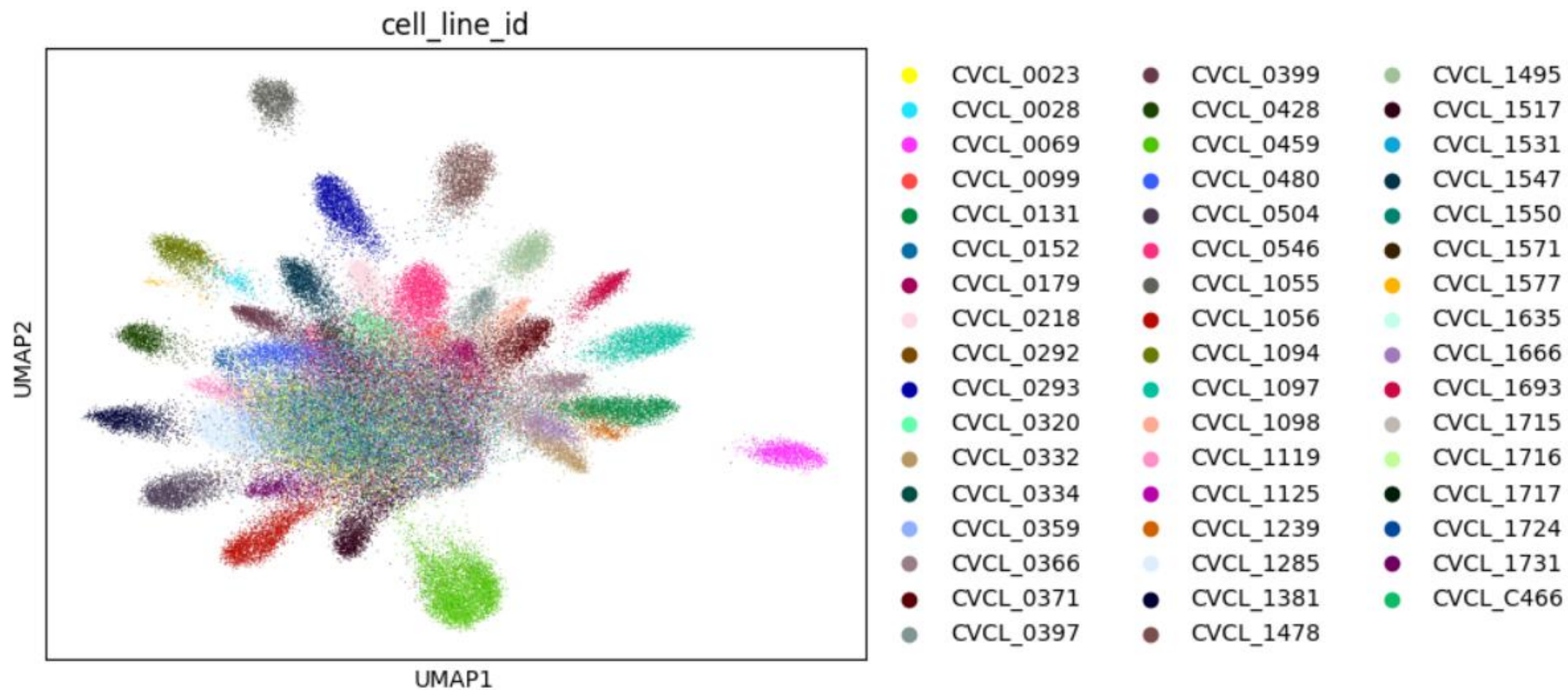
Solution:

- Only 100 cell for each drug-cell line – plate.
- Only 100 Monte Carlo samples
- The computation was divided in 17 parts so that is manageable for the clusters resources

Dimensionality reduction of differential gene-set scores

- The top 50 principal components of the differential gene-set matrix were used as input to cuML t-SNE for two-dimensional visualization.
- Neural Network Multidimensional Scaling (NNMDS) was trained on the first 128 principal components of gene-set scores to generate an alternative two-dimensional embedding.



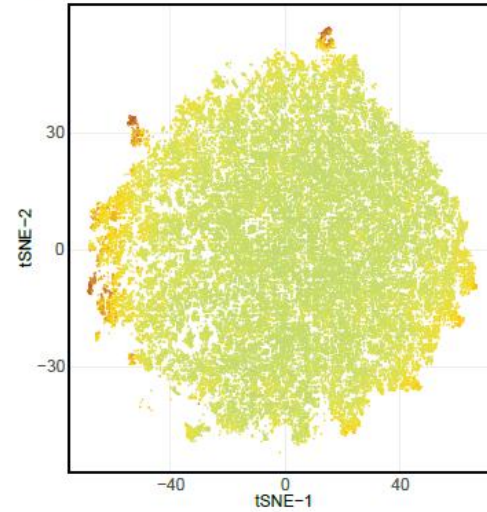


Next Step

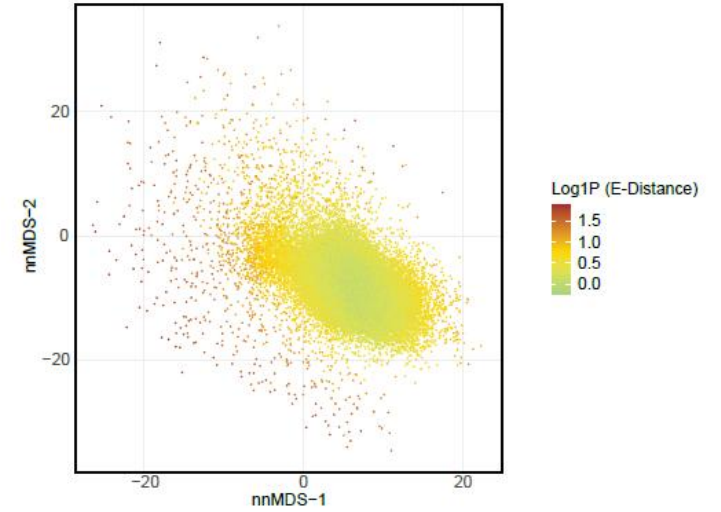
- Finish the differential expression pipeline in order to look like this:

E-distance: quantifies the separability of a perturbed population from its controls

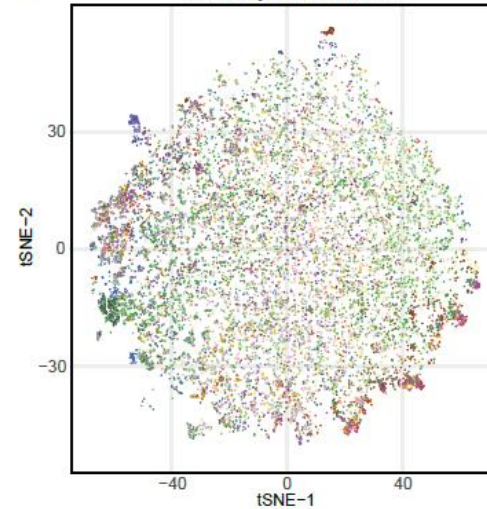
B TSNE Visualization of Differential Expression Space, Colored by E-Distance



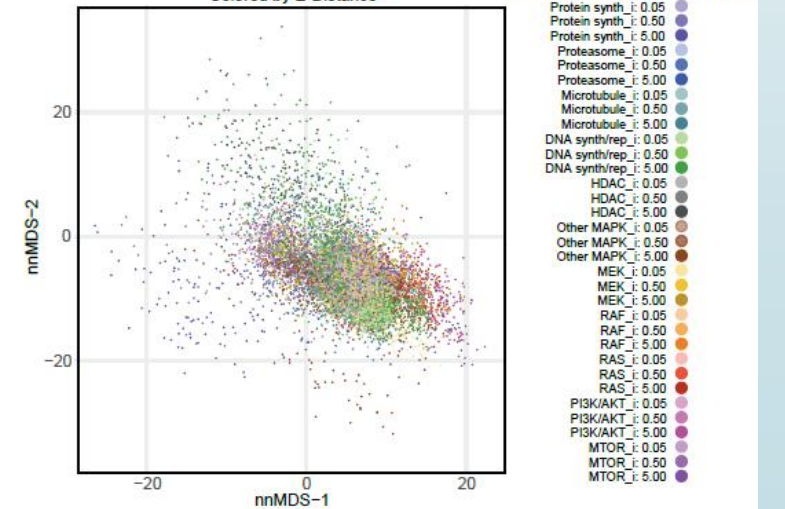
C nnMDS-2 Visualization of Differential Expression Space, Colored by E-Distance



D TSNE Visualization of Differential Expression Space, Colored by MOA & Distance



E nnMDS Visualization of Differential Expression Space, Colored by E-Distance





THANK YOU FOR YOUR ATTENTION